

$$(b) \frac{2}{3} + \frac{2}{9} + \dots + \frac{2}{3^n} = 1 - \frac{1}{3^n}, n \geq 1 \quad n = 1, 0, 2$$

$$n=1 \quad \frac{2}{3^1} = 1 - \frac{1}{3^1} \rightarrow \frac{2}{3} = \frac{2}{3} \quad \checkmark$$

$$n=p \quad \frac{2}{3} + \frac{2}{9} + \dots + \frac{2}{3^p} = 1 - \frac{1}{3^p}, \text{ H.I.}$$

$$n=p+1 \quad \frac{2}{3} + \frac{2}{9} + \dots + \frac{2}{3^p} + \frac{2}{3^{p+1}} = 1 - \frac{1}{3^{p+1}}$$

$$1 - \frac{1}{3^{p+1}}$$

$$\frac{3^{p+1} - 1}{3^{p+1}}$$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$\frac{(a+b)^2 + 1}{3}$$

$$\frac{a^2 + 2ab + b^2 + 1}{3}$$

Demonstracion

$$\frac{2}{3} + \frac{2}{9} + \dots + \frac{2}{3^p} + \frac{2}{3^{p+2}}$$

$$\frac{a}{b} + \frac{c}{b} = \frac{a+c}{b}$$

$$1 - \frac{1}{3^p} + \frac{2}{3^{p+2}}$$

$$\frac{a}{b} + c \cdot \frac{b}{b}$$

$$1 + \frac{2}{3^{p+2}} - \frac{1}{3^p} \cdot \frac{3^2}{3^2}$$

$$\frac{a+cb}{b}$$

$$1 + \frac{2}{3^{p+2}} - \frac{1 \cdot 3}{3^p \cdot 3^2}$$

$$3^{p+2} = 3^p \cdot 3^2$$

$$1 + \frac{2}{3^{p+2}} - \frac{3}{3^{p+2}}$$

$$\frac{a}{b} \rightarrow \frac{c}{d} = \frac{ac}{bd}$$

$$1 + \frac{2-3}{3^{p+2}}$$

$$1 - \frac{1}{3^{p+2}}$$

$$1. \sum_{i=0}^n (7-11i) = \frac{(n+1)(14-11n)}{2}$$

$$n=0 \quad 7-11 \cdot 0 = \frac{(1+0)(14-11 \cdot 0)}{2}$$

$$7=7 \checkmark$$

$$n=p \quad \sum_{i=0}^p (7-11i) = \frac{(p+1)(14-11p)}{2} \text{ H.I.}$$

$$n=p+1 \quad \sum_{i=0}^{p+1} (7-11i) = \frac{(p+1+1)(14-11(p+1))}{2}$$

$$\sum_{i=k}^{p+1} 1 = \sum_{i=k}^p 1 + (p+1)$$

$$\sum_{i=k}^{p+1} \frac{1}{j} = \sum_{i=k}^p \frac{1}{j} + \frac{1}{p+1}$$

$$\frac{(p+1+1)(14-11(p+1))}{2}$$

$$\frac{(p+2)(14-11p-11)}{2}$$

$$\boxed{\frac{(p+2)(3-11p)}{2}} \leftarrow \text{H.Q.D.}$$

$$p+2$$

$$\sum_{i=0}^p (7-11i) = \frac{(p+2)(3-11p)}{2}$$

$$i=0$$

$$2$$

$$p$$

$$\sum_{i=0}^p (7-11i) + (7-11(p+2))$$

$$i=0$$

$$7-11p-11$$

$$-11p-4$$

$$\frac{(p+2)(14-11p)}{2} + \frac{-11p-4}{2}$$

$$\frac{(p+2)(14-11p) + (-22p-8)}{2}$$

$$\frac{14p - 11p^2 + 14 - 11p - 22p - 8}{2}$$

$$\frac{-11p^2 - 19p + 6}{2}$$

$$\frac{(p+2)(3-11p)}{2}$$

$$-11p^2 - 19p + 6$$

$$\begin{aligned} (-11p + 3) + 3 &= 3p \\ (p + 2) + 2 &= -22p \\ &= -19p \end{aligned}$$

$$\frac{(-11p+3)(p+2)}{2}$$

$$\frac{(p+2)(3-11p)}{2}$$

$$x^2 + 2x + 1$$

$$x \cdot x = x^2$$

$$\begin{array}{rcl} x & \times & 1 = x \\ x & \times & 1 = \underline{x} \\ & & 2x \end{array}$$

$$(x+1)(x+1) = (x+1)^2$$

$$2(x+2)^2 + 3 \cdot (x+2)$$

$$(x+2)(2(x+2) + 3)$$

$$\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$$

$$x^a \cdot x^b = x^{a+b}$$

$$\frac{x^a}{x^b} = x^{a-b}$$

$$4. \sum_{i=0}^n \left(\frac{2}{3}\right)^i = 3 - \frac{2^{n+1}}{3^n} \quad a^0 = 1$$

$$n=0 \quad \left(\frac{2}{3}\right)^0 = 3 - \frac{2^{0+1}}{3^0}$$

$$1 = 1 \checkmark$$

$$n=p \quad \sum_{i=0}^p \left(\frac{2}{3}\right)^i = 3 - \frac{2^{p+1}}{3^p} \quad , \text{H.I.}$$

$$n=p+1 \quad \sum_{i=0}^{p+1} \left(\frac{2}{3}\right)^i = 3 - \frac{2^{(p+1)+1}}{3^{p+1}}$$

$$3 - \frac{2^{p+2}}{3^{p+1}} \quad , \text{H. Q. D.}$$

Demonstracion

$$\sum_{i=0}^{p+1} \left(\frac{2}{3}\right)^i$$

$$\sum_{i=0}^p \left(\frac{2}{3}\right)^i + \left(\frac{2}{3}\right)^{p+1}$$

$$3 - \frac{2^{p+1}}{3^p} + \left(\frac{2}{3}\right)^{p+1}$$

$$3 - \frac{2^{p+2}}{3^{p+1}}$$

$$3 - \frac{2^{p+1}}{3^p} + \left(\frac{2}{3} \right)^{p+1}$$

$$3 + \left(\frac{2}{3} \right)^{p+1} - \frac{2^{p+1}}{3^p}$$

$$\frac{2^{p+1}}{3^{p+1}} = \left(\frac{2}{3} \right)^{p+1}$$

$$\left(\frac{a}{b} \right)^x = \frac{a^x}{b^x}$$

$$3 + \frac{2^{p+1}}{3^{p+1}} - \frac{2^{p+1}}{3^p} \cdot \frac{3}{3}$$

$$3 - \frac{2^{p+2}}{3^{p+1}}$$

$$3 + \frac{2^{p+1}}{3^{p+1}} - \frac{2^{p+1}}{3^p} \cdot \frac{3}{3} \rightarrow 3^p \cdot \frac{3}{3} = 3^{p+1}$$

$$3 + \frac{2^{p+1}}{3^{p+1}} - \frac{2^{p+1} \cdot 3}{3^{p+1}}$$

$$3 + \frac{2^{p+1}}{3^{p+1}} - \frac{2^{p+1} \cdot 3}{3^{p+1}}$$

$$3 - \frac{2^{p+2}}{3^{p+1}}$$

$$3 + \frac{2^{p+1}}{3^{p+1}} (1 - 3)$$

$$2x + x$$

$$x \left(\frac{2x}{x} + \frac{x}{x} \right)$$

$$3 + \frac{2^{p+1} \cdot -2^1}{3^{p+1}}$$

$$x(2 + 1)$$

$$\boxed{3 - \frac{2^{p+2}}{3^{p+1}}}$$

$$-x^a \cdot x^b$$

$$-x^{a+b}$$

$$(c) 1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}, n \geq 1.$$

$$n=1 \quad 1^3 = \frac{1(1+1)^2}{4}$$

$$1=1 \quad \checkmark$$

$$n=p \quad 1^3 + 2^3 + 3^3 + \dots + p^3 = \frac{p^2(p+1)^2}{4} \text{ H.I.} \quad (p+1+1)^2$$

$$n=p+1 \quad 1^3 + 2^3 + 3^3 + \dots + p^3 + (p+1)^3 = \frac{(p+1)^2 \cdot (p+2)^2}{4}$$

$$1^3 + 2^3 + 3^3 + \dots + p^3 + (p+1)^3$$

$$\frac{p^2(p+1)^2}{4} + (p+1)^3 \quad \frac{4}{4}$$

$$\frac{p^2(p+1)^2 + (p+1)^3 \cdot 4}{4}$$

$$\frac{(p+1)^2 (p^2 + (p+1) \cdot 4)}{4}$$

$$\frac{(p+1)^2 (p^2 + 4p + 4)}{4}$$

$$\frac{(p+1)^2 \cdot (p+2)^2}{4}$$

$$\boxed{\frac{(p+1)^2 (p+2)^2}{4}}$$

$$p^2 + 4p + 4$$

$$\frac{(p+2)(p+2)}{(p+2)^2}$$

$$\begin{array}{r} (p+2) \\ + \\ (p+2) \\ \hline 2p+4 \end{array} \quad \begin{array}{r} (p+2) \\ + \\ (p+2) \\ \hline 2p+4 \end{array} \quad \frac{2p+4}{4p}$$